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from scipy import stats

# 1. Filter for rows where BOTH GRE and GRE V are non-missing and non-zero
correlation_data_1 = df.dropna(subset=['GRE', 'GRE V'])
correlation_data_1 = correlation_data_1[(correlation_data_1['GRE'] > 0) & (correlation_data_1['GRE V'] > 0)]

# 2. Compute Pearson Correlation
r_val, p_val_1 = stats.pearsonr(correlation_data_1['GRE'], correlation_data_1['GRE V'])

print("--- Pearson Correlation: GRE vs. GRE V ---")
print(f"Correlation Coefficient (r): {r_val:.4f}")
print(f"P-value: {p_val_1:.4e}")

--- Pearson Correlation: GRE vs. GRE V ---
Correlation Coefficient (r): 0.0764
P-value: 1.8889e-05

# 1. Filter for rows where BOTH GPA and GRE are non-missing and non-zero
correlation_data_2 = df.dropna(subset=['GPA', 'GRE'])
correlation_data_2 = correlation_data_2[(correlation_data_2['GPA'] > 0) & (correlation_data_2['GRE'] > 0)]

# 2. Compute Spearman Correlation
rho_val, p_val_2 = stats.spearmanr(correlation_data_2['GPA'], correlation_data_2['GRE'])

print("\n--- Spearman Correlation: GPA vs. GRE ---")
print(f"Correlation Coefficient (rho): {rho_val:.4f}")
print(f"P-value: {p_val_2:.4e}")

--- Spearman Correlation: GPA vs. GRE ---
Correlation Coefficient (rho): 0.0685
P-value: 7.1066e-05

```

A. Correlation analysis

Interpretation: GRE vs. GRE V (Pearson)

The Pearson correlation coefficient of **0.0764** indicates an extremely weak positive linear relationship between total GRE and GRE Verbal scores. Although the p-value () is well below the 0.05 threshold—meaning this result is unlikely to be due to random noise—the correlation is so low that a change in total GRE score provides almost no predictive insight into a student's Verbal score. This suggests that in this specific pool of applicants, total scores may be heavily influenced by other components, such as the Quantitative section.

Interpretation: GPA vs. GRE (Spearman)

The Spearman correlation of **0.0685** reveals a negligible monotonic relationship between undergraduate GPA and total GRE scores. While the result is statistically significant (), the coefficient is near zero, implying that a student's long-term academic performance (GPA) and their performance on a single standardized test (GRE) are nearly independent of one another. For admissions analytics, this suggests that these two metrics capture entirely different, unrelated aspects of an applicant's profile.

--- Group Comparison: GPA of Accepted vs. Rejected ---

Test Used: Independent T-Test (Welch's)

Test Statistic: -4.1310

P-value: 3.6271e-05

Mean GPA (Accepted): 3.744

Mean GPA (Rejected): 3.760

--- Categorical Association: Degree vs. US/International ---

Contingency Table:

US/International	American	International
Degree		
Master's	4875	4921
PhD	15579	16902
PsyD	686	83

Chi-Square Statistic: 513.4690

P-value: 3.1741e-112

Degrees of Freedom: 2

B. Group Comparison: GPA by Outcome

Interpretation: The independent t-test yielded a p-value of , which is significantly lower than the standard threshold, indicating a statistically significant difference between the two groups. However, the interpretation is surprising: the mean GPA for rejected applicants () is actually slightly **higher** than that of accepted applicants (). This suggests that for this specific dataset, a higher GPA does not guarantee admission and that other factors—such as research experience, letters of recommendation, or statement of purpose—likely carry heavier weight in the final decision.

C. Categorical Association: Degree vs. US/International

Interpretation: The Chi-Square test resulted in an extremely low p-value of , providing overwhelming evidence that the type of degree sought is not independent of an applicant's citizenship status. Looking at the contingency table, we can see clear patterns: while PhD applications are fairly balanced between American and International students, the **PsyD** program is heavily dominated by American applicants (vs.). This strong association suggests that domestic and international students have significantly different preferences or requirements when choosing between professional doctorates (PsyD) and research doctorates (PhD).

Analytical Summary: Graduate Admissions Data Study

1. PhD Applicant Profile (GRE)

The average GRE score for an applicant accepted into a PhD program is [Insert Value].

- **How to get this:** Run `df[(df['Degree'] == 'PhD') & (df['outcome'] == 'Accepted')]['GRE'].mean()` in your notebook.
- **Interpretation:** This score serves as a benchmark for competitive research-based programs. However, its utility is limited by the fact that many programs have made the GRE optional, leading to potential "self-selection bias" where only high-scoring students report their results.

2. Master's Applicant Profile (GPA)

The average GPA for an applicant accepted into a Master's program is [Insert Value].

- **How to get this:** Run `df[(df['Degree'] == "Master's") & (df['outcome'] == 'Accepted')]['GPA'].mean()` in your notebook.
- **Interpretation:** While this average suggests a high standard of academic consistency, the GPA metric does not account for the "prestige" or rigor of the undergraduate institution, which is a significant variable in admissions logic.

3. Shortcomings of Date Inference

Inferring a `decision_date` from a text field and a term value (e.g., "Fall 2024") has several limitations:

1. **Granularity Loss:** Since we assigned a "best-guess" year based on the term, we lose the exact day the decision was actually made, which may lead to artificial clusters in our timeline.
2. **Year Misalignment:** Applicants often post decisions from previous cycles or "late" entries for spring terms, which can lead to data being assigned to the wrong calendar year.
3. **Ambiguity:** Status fields often contain "Waitlisted" or "Interview" dates rather than final "Accepted/Rejected" dates, creating noise in the `days_to_decision` calculation.

4. Correlation Analysis (GPA vs. GRE)

Our correlation analysis (Spearman's ρ) suggests a **negligible relationship** between undergraduate GPA and GRE scores. Despite being statistically significant due to the large sample size, the coefficient is near zero. This implies that academic performance over four years (GPA) is independent of standardized testing performance (GRE); they represent two distinct facets of an applicant's profile.

5. Group Comparison (Accepted vs. Rejected GPA)

The independent t-test revealed a statistically significant difference (), but surprisingly, the mean GPA of **rejected** applicants () was slightly higher than that of **accepted** applicants (). This suggests that GPA is a "threshold" metric rather than a deciding factor; once a student meets a certain academic bar, other qualitative factors like research fit and letters of intent likely determine the final outcome.

6. Categorical Association (Degree vs. US/International)

The Chi-Square test indicated a strong association between degree type and citizenship status (). The data shows that while PhD applications are common across both groups, **PsyD** programs are predominantly sought by domestic American students. This suggests that international students may prioritize research-focused degrees (PhD) over professional practice degrees (PsyD) due to funding availability or global recognition.

7. Remaining Data Quality Issues

Even after extensive cleaning, several issues remain:

- **Self-Reporting Bias:** All data is user-submitted (The Grad Cafe), meaning users may exaggerate scores or only post if they feel successful.
- **Missingness:** Large portions of the dataset still contain `NaN` values for GRE and GPA, which can skew averages if the "missing" group is systematically different from the "reporting" group.
- **Normalization:** GPA values were cleaned, but they were not normalized against different grading scales (e.g., International 10-point vs. US 4.0 scales), which introduces "noise" into the comparison.